

Chapter 1

Transit-Time Factor

1.1 Physical Motivation

A radio-frequency accelerating gap presents a longitudinal electric field that oscillates at angular frequency ω . Because the field reverses direction every half period, a particle of finite velocity cannot experience the peak field throughout its entire traversal of the gap. The effective energy gain is reduced relative to the instantaneous peak voltage by a factor that depends on how long the particle spends inside the gap compared to the RF period. This reduction is quantified by the **transit-time factor** T , a dimensionless quantity ($0 \leq T \leq 1$) determined by the spatial field profile and the particle trajectory through the gap.

1.2 Problem Setup

Consider an accelerating gap. The on-axis longitudinal electric field is

$$E_z(s, t) = E(s) \cos(\omega t + \varphi_s + \Delta\varphi)$$

where $E(s) \equiv E(r = 0, s)$ is the spatial field profile (units: V/m), φ_s is the synchronous phase (a fixed design parameter), and $\Delta\varphi$ is the phase deviation of this particle from the synchronous particle. A particle of charge q traverses a region of length L containing the field. We choose the time origin $t = 0$ when the particle crosses $s = 0$, and measure positions relative to this origin so the field extends over $-L/2 \leq s \leq L/2$. The choice of where to place this origin within the gap is free — its optimal location will emerge naturally from the derivation.

The particle arrives at position s at time

$$t(s) = \int_0^s \frac{ds'}{v(s')}$$

where $v(s)$ is the (possibly varying) particle velocity. No assumption about constant velocity is made at this stage.

1.3 Notation

Symbol	Meaning
ΔU	Total energy gain of a particle crossing one gap

Symbol	Meaning
ΔW	Energy deviation from the synchronous particle (longitudinal coordinate)
φ_s	Synchronous phase (design parameter)
$\Delta\varphi$	Phase deviation from synchronous (longitudinal coordinate)

The two are related: after computing ΔU for both the synchronous particle ($\Delta\varphi = 0$) and an arbitrary particle, the change in the deviation variable per gap is $\delta(\Delta W) = \Delta U - \Delta U_s = qV_0T[\cos(\varphi_s + \Delta\varphi) - \cos\varphi_s]$.

1.4 General Energy-Gain Integral

The energy gained by the particle crossing the gap is the work done by the longitudinal field along its trajectory:

$$\Delta U = q \int_{-L/2}^{L/2} E(s) \cos(\omega t(s) + \varphi_s + \Delta\varphi) ds$$

1.5 Trigonometric Expansion (Exact)

Expand the cosine using the angle-addition identity:

$$\cos(\omega t(s) + \varphi_s + \Delta\varphi) = \cos(\omega t(s)) \cos(\varphi_s + \Delta\varphi) - \sin(\omega t(s)) \sin(\varphi_s + \Delta\varphi)$$

The energy gain separates into two terms:

$$\Delta U = q \cos(\varphi_s + \Delta\varphi) \int_{-L/2}^{L/2} E(s) \cos(\omega t(s)) ds - q \sin(\varphi_s + \Delta\varphi) \int_{-L/2}^{L/2} E(s) \sin(\omega t(s)) ds$$

1.6 Factored Form and General Transit-Time Factor

Define the **gap voltage** (DC voltage):

$$V_0 \equiv \int_{-L/2}^{L/2} E(s) ds$$

This is the voltage a particle would gain if it traversed the gap instantaneously (the electrostatic limit). We also introduce shorthand for the two integrals:

$$I_c \equiv \int_{-L/2}^{L/2} E(s) \cos(\omega t(s)) ds, \quad I_s \equiv \int_{-L/2}^{L/2} E(s) \sin(\omega t(s)) ds$$

With these definitions the energy gain from the previous section reads

$$\Delta U = q \cos(\varphi_s + \Delta\varphi) I_c - q \sin(\varphi_s + \Delta\varphi) I_s$$

We want to write this in the factored form $\Delta U = qV_0T \cos(\varphi_s + \Delta\varphi)$. Dividing both sides by $q \cos(\varphi_s + \Delta\varphi)$:

$$V_0 T = I_c - \tan(\varphi_s + \Delta\varphi) I_s$$

Solving for T :

$$T \equiv \frac{I_c}{V_0} - \tan(\varphi_s + \Delta\varphi) \frac{I_s}{V_0}$$

or, written out explicitly (following Wangler [1]):

$$T \equiv \frac{\int_{-L/2}^{L/2} E(s) \cos(\omega t(s)) ds}{\int_{-L/2}^{L/2} E(s) ds} - \tan(\varphi_s + \Delta\varphi) \frac{\int_{-L/2}^{L/2} E(s) \sin(\omega t(s)) ds}{\int_{-L/2}^{L/2} E(s) ds}$$

This is exact but unsatisfying: T depends on the phase through the I_s term. The dependence arises from the choice of time origin — i.e., where we place $s = 0$ within the gap. If we shift the origin, $t(s)$ changes, and the balance between I_c and I_s shifts.

1.7 The Electrical Center

We can eliminate the phase-dependence of T by choosing the origin such that $I_s = 0$. Define the **electrical center** of the gap as the point $s = 0$ where

$$I_s = \int_{-L/2}^{L/2} E(s) \sin(\omega t(s)) ds = 0$$

When the origin is placed at the electrical center, the $\tan(\varphi_s + \Delta\varphi)$ term vanishes and the transit-time factor becomes purely the cosine-weighted ratio:

$$T = \frac{\int_{-L/2}^{L/2} E(s) \cos(\omega t(s)) ds}{\int_{-L/2}^{L/2} E(s) ds}$$

and the energy gain takes the clean factored form:

$$\Delta U = q V_0 T \cos(\varphi_s + \Delta\varphi)$$

This result is **exact** — it requires only that the origin be placed at the electrical center, with no assumption about constant velocity.

When does the electrical center coincide with the geometric center? If $E(s)$ is symmetric about the geometric center and the velocity is constant during transit, then $t(s) = s/v$ is odd, $\sin(\omega t(s))$ is odd, and the product $E(s) \sin(\omega t(s))$ is odd — so the sine integral vanishes by symmetry. In this case the geometric and electrical centers coincide.

1.8 Constant-Velocity Approximation

In most practical cases the fractional energy change across a single gap is small compared to the particle's kinetic energy ($\Delta U/W_{\text{kin}} \ll 1$), so the particle velocity is approximately constant during transit. Setting $t(s) = s/v$ the transit-time factor becomes

$$T = \frac{\int_{-L/2}^{L/2} E(s) \cos\left(\frac{\omega s}{v}\right) ds}{\int_{-L/2}^{L/2} E(s) ds}$$

This is the form most commonly encountered in the literature. The constant-velocity approximation enters only when we wish to **evaluate** T — the factored energy-gain formula $\Delta U = qV_0 T \cos(\varphi_s + \Delta\varphi)$ itself is exact.

1.9 Connection to Longitudinal Dynamics

For the **synchronous particle** ($\Delta\varphi = 0$), the energy gain per gap is

$$\Delta U_s = qV_0 T \cos \varphi_s$$

For an arbitrary particle the energy gain is $\Delta U = qV_0 T \cos(\varphi_s + \Delta\varphi)$. The change in the energy-deviation coordinate $\Delta W = W - W_s$ per gap is therefore

$$\delta(\Delta W) = \Delta U - \Delta U_s = qV_0 T [\cos(\varphi_s + \Delta\varphi) - \cos \varphi_s]$$

which is precisely the right-hand side of the longitudinal equation of motion.

1.10 Worked Example: Uniform Field

Take the simplest model — a constant field across the gap with hard edges:

$$E(s) = \begin{cases} E_0 & |s| \leq L/2 \\ 0 & |s| > L/2 \end{cases}$$

The field is symmetric, so with $t(s) = s/v$ the geometric center is the electrical center.

Gap voltage:

$$V_0 = E_0 L$$

Transit-time factor:

$$T = \frac{\int_{-L/2}^{L/2} E_0 \cos\left(\frac{\omega s}{v}\right) ds}{E_0 L} = \frac{1}{L} \int_{-L/2}^{L/2} \cos\left(\frac{\omega s}{v}\right) ds$$

Evaluating the integral:

$$\int_{-L/2}^{L/2} \cos\left(\frac{\omega s}{v}\right) ds = \frac{v}{\omega} \left[\sin\left(\frac{\omega s}{v}\right) \right]_{-L/2}^{L/2} = \frac{v}{\omega} \left[\sin\left(\frac{\omega L}{2v}\right) - \sin\left(-\frac{\omega L}{2v}\right) \right] = \frac{2v}{\omega} \sin\left(\frac{\omega L}{2v}\right)$$

where in the last step we used $\sin(-x) = -\sin(x)$. Therefore

$$T = \frac{1}{L} \cdot \frac{2v}{\omega} \sin\left(\frac{\omega L}{2v}\right)$$

Define the **transit angle** $\theta = \omega L/v$, the total RF phase advance during one traversal of the gap. Then $\omega/v = \theta/L$ and $\omega L/(2v) = \theta/2$, so

$$T = \frac{2v}{\omega L} \sin\left(\frac{\theta}{2}\right) = \frac{2}{\theta} \sin\left(\frac{\theta}{2}\right) = \frac{\sin(\theta/2)}{\theta/2}$$

This is a sinc function of the half transit angle.

1.11 Limiting Cases and Physical Discussion

Thin gap ($\theta \rightarrow 0$): Taylor-expanding, $\sin(\theta/2) \approx \theta/2 - (\theta/2)^3/6 + \dots$, so $T \rightarrow 1$. The particle traverses the gap in a time much shorter than the RF period and sees an essentially static field.

Half-wavelength cell ($L = \beta\lambda/2$, so $\theta = \pi$): This is the standard cell length for a π -mode standing-wave structure operating at synchronous velocity $v = \beta c$. Here

$$T = \frac{\sin(\pi/2)}{\pi/2} = \frac{2}{\pi} \approx 0.637$$

Even at the optimal cell length, approximately 36% of the peak voltage is lost to transit-time effects when the field is uniform.

Design trade-off: The transit-time factor encodes a fundamental tension in cavity design:

- **Shorter gaps** ($\theta \ll 1$) give $T \approx 1$, but the achievable voltage $V_0 = E_0 L$ is limited by the breakdown field E_0 .
- **Longer gaps** support higher V_0 but reduce T , so the effective acceleration $V_0 T$ eventually saturates or decreases.

In practice, cavity designers shape the field profile to concentrate the field near the gap center — using **nose cones** or reentrant geometry — thereby increasing T beyond the uniform-field value at the same physical gap length L , at the cost of higher peak surface fields.

1.12 References

- [1] T.P. Wangler, *RF Linear Accelerators*, 2nd ed. (Wiley-VCH, 2008), Sec. 2.2.
- [2] S.Y. Lee, *Accelerator Physics*, 3rd ed. (World Scientific, 2012), Ch. 3, Sec. VIII.2
- [3] H. Wiedemann, *Particle Accelerator Physics*, 4th ed. (Springer, 2015), Ch. 18, Sec. 18.2.3

Chapter 2

Longitudinal Equations of Motion

2.1 Introduction

In the previous note on the transit-time factor, we derived the energy gain of a single particle crossing one accelerating gap:

$$\Delta U = qV_0 T \cos(\varphi_s + \Delta\varphi)$$

and the per-gap change in the energy deviation:

$$\delta(\Delta W) = qV_0 T [\cos(\varphi_s + \Delta\varphi) - \cos \varphi_s]$$

We now develop the complete longitudinal dynamics: how the phase deviation $\Delta\varphi$ and energy deviation ΔW evolve as a particle traverses many accelerating cells. Starting from discrete difference equations (one update per gap), we pass to the continuum limit to obtain coupled differential equations of motion, and finally construct the Hamiltonian that generates them.

2.2 Difference Equations (Per Cell)

Consider a linac composed of identical cells of length L_{cell} . Each cell contains one accelerating gap. Number the cells by index $n = 0, 1, 2, \dots$. The synchronous particle arrives at gap n with the design phase φ_s and the design energy W_s ; an arbitrary particle arrives with phase $\varphi_s + \Delta\varphi_n$ and energy $W_s + \Delta W_n$.

2.2.1 Energy kick

The energy deviation changes at each gap according to the result from the transit-time factor derivation. Since $V_0 = E_0 L_{\text{cell}}$ (the gap voltage equals the average axial field times the cell length), the energy kick is

$$\boxed{\delta(\Delta W)_n = qE_0 T L_{\text{cell}} [\cos(\varphi_s + \Delta\varphi_n) - \cos \varphi_s]}$$

This is a thin-lens kick: the energy changes discontinuously at the gap, and the particle drifts between gaps with constant energy.

2.2.2 Phase slip

Between gap n and gap $n + 1$, a particle with the wrong energy arrives at a different time — and therefore a different RF phase — than the synchronous particle. We now derive this phase slip carefully.

Step 1: Transit times. The synchronous particle traverses one cell of length L_{cell} at velocity $v_s = \beta_s c$, taking time

$$\Delta t_s = \frac{L_{\text{cell}}}{\beta_s c}$$

An off-energy particle with velocity $v = \beta c$ traverses the same cell in

$$\Delta t = \frac{L_{\text{cell}}}{\beta c}$$

Step 2: Time difference. The arrival-time difference at gap $n + 1$ is

$$\Delta t - \Delta t_s = \frac{L_{\text{cell}}}{c} \left(\frac{1}{\beta} - \frac{1}{\beta_s} \right)$$

Step 3: Convert to phase difference. The RF phase at a fixed location advances as $\varphi_{\text{RF}}(t) = \omega t + \varphi_0$. If the off-energy particle arrives at the next gap a time $(\Delta t - \Delta t_s)$ later than the synchronous particle, it encounters the RF at a correspondingly more advanced phase. The phase slip accumulated over one cell is therefore

$$\delta(\Delta\varphi)_n = \omega(\Delta t - \Delta t_s) = \frac{2\pi}{\lambda} L_{\text{cell}} \left(\frac{1}{\beta} - \frac{1}{\beta_s} \right)$$

where $\omega = 2\pi c/\lambda$. Sign check: a faster particle ($\beta > \beta_s$) has $1/\beta < 1/\beta_s$, so $\delta(\Delta\varphi) < 0$ — it arrives early and sees a less advanced RF phase.

Step 4: Linearize the velocity difference. We express $(1/\beta - 1/\beta_s)$ in terms of the energy deviation ΔW in three sub-steps.

Step 4a — Relate $\Delta\beta$ to ΔW . Start from the Lorentz factor $\gamma = (1 - \beta^2)^{-1/2}$. Differentiating:

$$d\gamma = \gamma^3 \beta d\beta$$

Since both particles share the same rest mass, the energy deviation satisfies $\Delta W = m_0 c^2 \Delta\gamma$ regardless of whether W denotes total or kinetic energy — the rest-mass contribution cancels in the difference. Combining:

$$\Delta\beta = \frac{\Delta\gamma}{\gamma_s^3 \beta_s} = \frac{\Delta W}{m_0 c^2 \gamma_s^3 \beta_s}$$

Step 4b — Taylor-expand $1/\beta$. Write $\beta = \beta_s + \Delta\beta$ and expand to first order:

$$\frac{1}{\beta} = \frac{1}{\beta_s + \Delta\beta} \approx \frac{1}{\beta_s} \left(1 - \frac{\Delta\beta}{\beta_s} \right)$$

Therefore:

$$\frac{1}{\beta} - \frac{1}{\beta_s} \approx -\frac{\Delta\beta}{\beta_s^2}$$

Step 4c — Substitute. Inserting the result of Step 4a:

$$\frac{1}{\beta} - \frac{1}{\beta_s} \approx -\frac{1}{\beta_s^2} \cdot \frac{\Delta W}{m_0 c^2 \gamma_s^3 \beta_s} = -\frac{\Delta W}{m_0 c^2 \beta_s^3 \gamma_s^3}$$

This is the key kinematic identity:

$$\boxed{\frac{1}{\beta} - \frac{1}{\beta_s} \approx -\frac{\Delta W}{m_0 c^2 \beta_s^3 \gamma_s^3}}$$

Step 5: Assemble the phase-slip equation. Substituting the linearized result into the Step 3 expression:

$$\delta(\Delta\varphi)_n = \frac{2\pi L_{\text{cell}}}{\lambda} \cdot \left(-\frac{\Delta W_n}{m_0 c^2 \beta_s^3 \gamma_s^3} \right)$$

$$\boxed{\delta(\Delta\varphi)_n = -\frac{2\pi L_{\text{cell}}}{\lambda m_0 c^2 \beta_s^3 \gamma_s^3} \Delta W_n}$$

Consistent with our sign check: $\Delta W > 0$ gives $\delta(\Delta\varphi) < 0$.

2.2.3 Summary of difference equations

$$\delta(\Delta\varphi)_n = -\frac{2\pi L_{\text{cell}}}{\lambda m_0 c^2 \beta_s^3 \gamma_s^3} \Delta W_n$$

$$\delta(\Delta W)_n = qE_0 T L_{\text{cell}} [\cos(\varphi_s + \Delta\varphi_n) - \cos \varphi_s]$$

These form a symplectic map: the energy is kicked at the gap, then the phase drifts in the inter-gap space.

2.3 Conversion to Differential Equations

2.3.1 From difference to derivative in n

When $\Delta\varphi$ and ΔW change by small amounts per cell — the **adiabatic condition** — we may replace the finite differences by derivatives:

$$\delta(\Delta\varphi)_n \longrightarrow \frac{d\Delta\varphi}{dn}, \quad \delta(\Delta W)_n \longrightarrow \frac{d\Delta W}{dn}$$

This gives

$$\frac{d\Delta\varphi}{dn} = -\frac{2\pi L_{\text{cell}}}{\lambda m_0 c^2 \beta_s^3 \gamma_s^3} \Delta W$$

$$\frac{d\Delta W}{dn} = qE_0 T L_{\text{cell}} [\cos(\varphi_s + \Delta\varphi) - \cos \varphi_s]$$

The validity of this smooth approximation requires that the change per cell be small compared to the oscillation amplitude — equivalently, that the synchrotron oscillation period spans many cells.

2.3.2 From n to s

The independent variable n counts cells; the distance along the linac is s . Since each cell has length L_{cell} :

$$ds = L_{\text{cell}} dn \quad \implies \quad \frac{d}{ds} = \frac{1}{L_{\text{cell}}} \frac{d}{dn}$$

Dividing both equations by L_{cell} :

$$\begin{aligned} \frac{d\Delta\varphi}{ds} &= \frac{1}{L_{\text{cell}}} \cdot \left(-\frac{2\pi L_{\text{cell}}}{\lambda m_0 c^2 \beta_s^3 \gamma_s^3} \Delta W \right) = -\frac{2\pi}{\lambda m_0 c^2 \beta_s^3 \gamma_s^3} \Delta W \\ \frac{d\Delta W}{ds} &= \frac{1}{L_{\text{cell}}} \cdot qE_0 T L_{\text{cell}} [\cos(\varphi_s + \Delta\varphi) - \cos \varphi_s] = qE_0 T [\cos(\varphi_s + \Delta\varphi) - \cos \varphi_s] \end{aligned}$$

The cell length L_{cell} cancels in both equations. This is why the final equations of motion are the same regardless of whether one uses $\beta_s \lambda$ cells (DTL, 2π -mode) or $\beta_s \lambda/2$ cells (π -mode standing wave): the difference equations scale with L_{cell} , but so does the step size ds , and the ratio is independent of cell length.

2.3.3 Longitudinal equations of motion

$$\boxed{\frac{d\Delta\varphi}{ds} = -\frac{2\pi}{\lambda m_0 c^2 \beta_s^3 \gamma_s^3} \Delta W}$$

$$\boxed{\frac{d\Delta W}{ds} = qE_0 T [\cos(\varphi_s + \Delta\varphi) - \cos \varphi_s]}$$

These are the fundamental coupled equations governing longitudinal beam dynamics in a linac.

2.4 The Longitudinal Hamiltonian

2.4.1 Canonical structure

The equations of motion have the structure of Hamilton's equations if we identify $(\Delta\varphi, \Delta W)$ as a canonical coordinate-momentum pair with s as the independent variable:

$$\frac{d\Delta\varphi}{ds} = \frac{\partial H}{\partial(\Delta W)}, \quad \frac{d\Delta W}{ds} = -\frac{\partial H}{\partial(\Delta\varphi)}$$

We construct $H(\Delta\varphi, \Delta W; s)$ by identifying kinetic and potential terms.

2.4.2 Kinetic term

From the first equation of motion:

$$\frac{\partial H}{\partial(\Delta W)} = -\frac{2\pi}{\lambda m_0 c^2 \beta_s^3 \gamma_s^3} \Delta W$$

Integrating with respect to ΔW :

$$K(\Delta W) = -\frac{2\pi}{\lambda m_0 c^2 \beta_s^3 \gamma_s^3} \cdot \frac{\Delta W^2}{2} = -\frac{\pi}{\lambda m_0 c^2 \beta_s^3 \gamma_s^3} \Delta W^2$$

2.4.3 Potential term

From the second equation of motion:

$$-\frac{\partial H}{\partial(\Delta\varphi)} = qE_0T[\cos(\varphi_s + \Delta\varphi) - \cos\varphi_s]$$

Therefore

$$\frac{\partial H}{\partial(\Delta\varphi)} = -qE_0T[\cos(\varphi_s + \Delta\varphi) - \cos\varphi_s]$$

Integrating with respect to $\Delta\varphi$:

$$V(\Delta\varphi) = -qE_0T \int_0^{\Delta\varphi} [\cos(\varphi_s + \varphi') - \cos\varphi_s] d\varphi'$$

Evaluating the integral:

$$\int_0^{\Delta\varphi} \cos(\varphi_s + \varphi') d\varphi' = \sin(\varphi_s + \Delta\varphi) - \sin\varphi_s$$

$$\int_0^{\Delta\varphi} \cos\varphi_s d\varphi' = \Delta\varphi \cos\varphi_s$$

Therefore

$$V(\Delta\varphi) = -qE_0T [\sin(\varphi_s + \Delta\varphi) - \sin\varphi_s - \Delta\varphi \cos\varphi_s]$$

The constant $-\sin\varphi_s$ shifts the zero of the potential but does not affect the equations of motion (it vanishes under $\partial/\partial(\Delta\varphi)$). We absorb it into the potential:

$$V(\Delta\varphi) = -qE_0T [\sin(\varphi_s + \Delta\varphi) - \Delta\varphi \cos\varphi_s] + qE_0T \sin\varphi_s$$

Since only derivatives of H enter the equations of motion, we are free to drop the additive constant $qE_0T \sin\varphi_s$. The conventional choice is

$$V(\Delta\varphi) = -qE_0T [\sin(\varphi_s + \Delta\varphi) - \Delta\varphi \cos\varphi_s]$$

2.4.4 The Hamiltonian

Combining kinetic and potential terms:

$$H = -\frac{\pi}{\lambda m_0 c^2 \beta_s^3 \gamma_s^3} \Delta W^2 - qE_0T [\sin(\varphi_s + \Delta\varphi) - \Delta\varphi \cos\varphi_s]$$

2.4.5 Verification

First Hamilton's equation:

$$\frac{\partial H}{\partial(\Delta W)} = -\frac{2\pi}{\lambda m_0 c^2 \beta_s^3 \gamma_s^3} \Delta W = \frac{d\Delta\varphi}{ds} \quad \checkmark$$

Second Hamilton's equation:

$$\begin{aligned} \frac{\partial H}{\partial(\Delta\varphi)} &= -qE_0 T [\cos(\varphi_s + \Delta\varphi) - \cos \varphi_s] \\ -\frac{\partial H}{\partial(\Delta\varphi)} &= qE_0 T [\cos(\varphi_s + \Delta\varphi) - \cos \varphi_s] = \frac{d\Delta W}{ds} \quad \checkmark \end{aligned}$$

Both Hamilton's equations reproduce the equations of motion.

2.5 Physical Interpretation

2.5.1 Fixed points

A fixed point is a location in phase space where a particle, once placed, remains indefinitely — all time derivatives vanish. Fixed points organize the global structure of the motion: nearby trajectories either orbit around them (stable) or diverge away from them (unstable).

Setting $d\Delta\varphi/ds = 0$ and $d\Delta W/ds = 0$ simultaneously requires:

- $\Delta W = 0$ (from the first equation)
- $\cos(\varphi_s + \Delta\varphi) = \cos \varphi_s$ (from the second equation)

The second condition has two solutions:

- $(\Delta\varphi, \Delta W) = (0, 0)$ — the **stable fixed point** (SFP). Nearby particles execute closed elliptical orbits around it. This is the synchronous particle itself, sitting at the center of the RF bucket.
- $(\Delta\varphi, \Delta W) = (-2\varphi_s, 0)$ — the **unstable fixed point** (UFP). Nearby trajectories are hyperbolas that diverge away. The separatrix passes through this point, dividing phase space into captured and uncaptured regions.

2.5.2 Phase stability

Linearizing the second equation of motion about the synchronous particle ($\Delta\varphi \ll 1$, $\Delta W \ll 1$):

$$\frac{d\Delta W}{ds} \approx -qE_0 T \sin \varphi_s \cdot \Delta\varphi$$

Combined with the first equation, this gives oscillatory (stable) solutions when the effective restoring force has the correct sign. Taking the second derivative:

$$\frac{d^2 \Delta\varphi}{ds^2} = -\frac{2\pi}{\lambda m_0 c^2 \beta_s^3 \gamma_s^3} \cdot \frac{d\Delta W}{ds} = \frac{2\pi q E_0 T \sin \varphi_s}{\lambda m_0 c^2 \beta_s^3 \gamma_s^3} \Delta\varphi$$

This is a simple harmonic oscillator if and only if the coefficient is negative:

$$\frac{2\pi q E_0 T \sin \varphi_s}{\lambda m_0 c^2 \beta_s^3 \gamma_s^3} < 0$$

For a positively charged particle ($q > 0$) in a linac where $E_0 T > 0$, stability requires

$$\sin \varphi_s < 0 \quad \Longleftrightarrow \quad -\pi < \varphi_s < 0$$

This is the **phase stability condition**: the synchronous particle must ride on the negative-slope side of the RF wave so that late particles see a stronger field and early particles see a weaker field — a restoring mechanism.

In addition, the synchronous particle must gain energy. The energy gain per gap is $qV_0 T \cos \varphi_s$, so net acceleration requires

$$\cos \varphi_s > 0 \quad \Longleftrightarrow \quad -\frac{\pi}{2} < \varphi_s < \frac{\pi}{2}$$

Combining both conditions gives the operating range for an accelerating linac:

$$-\frac{\pi}{2} < \varphi_s < 0$$

2.5.3 Separatrix

The Hamiltonian is a constant of motion (when β_s , γ_s , and $E_0 T$ are treated as slowly varying parameters). The separatrix is the phase-space trajectory that passes through the unstable fixed point $(\Delta\varphi, \Delta W) = (-2\varphi_s, 0)$. Setting $H(\Delta\varphi, \Delta W) = H(-2\varphi_s, 0)$:

$$-\frac{\pi}{\lambda m_0 c^2 \beta_s^3 \gamma_s^3} \Delta W^2 = qE_0 T [\sin(\varphi_s + \Delta\varphi) - \Delta\varphi \cos \varphi_s] - qE_0 T [\sin(-\varphi_s) + 2\varphi_s \cos \varphi_s]$$

Particles inside the separatrix execute bounded oscillations about the synchronous particle and are captured into the RF bucket; particles outside the separatrix are lost. The area enclosed by the separatrix defines the **longitudinal acceptance** of the linac.

2.6 Appendix: Cosine vs Sine Convention

The two conventions are related by a single phase shift:

$$\phi = \varphi + \frac{\pi}{2}$$

Quantity	Cosine (Wangler, these notes)	Sine (SY Lee)
Phase variable	φ	ϕ
Relation	—	$\phi = \varphi + \pi/2$
Energy gain	$\Delta W = qV_0 T \cos \varphi$	$\Delta W = qV_0 T \sin \phi$
On-crest phase	$\varphi = 0$	$\phi = \pi/2$
Zero-crossing phase	$\varphi = -\pi/2$	$\phi = 0$
Stable range (accel + focus)	$-\pi/2 < \varphi_s < 0$	$0 < \phi_s < \pi/2$
Phase stability condition	$\sin \varphi_s < 0$	$\cos \phi_s > 0$
Slope condition	$dV/d\varphi _{\varphi_s} > 0$	$dV/d\phi _{\phi_s} > 0$

Quantity	Cosine (Wangler, these notes)	Sine (SY Lee)
Synchrotron wavenumber	$k_s^2 \propto -\sin \varphi_s$	$k_s^2 \propto \cos \phi_s$
EOM: $d\Delta W/ds$	$qE_0T[\cos(\varphi_s + \Delta\varphi) - \cos \varphi_s]$	$qE_0T[\sin(\phi_s + \Delta\phi) - \sin \phi_s]$
Hamiltonian potential	$-qE_0T[\sin(\varphi_s + \Delta\varphi) - \Delta\varphi \cos \varphi_s]$	$qE_0T[\cos(\phi_s + \Delta\phi) + \Delta\phi \sin \phi_s]$
UFP location	$\Delta\varphi = -2\varphi_s$	$\Delta\phi = \pi - 2\phi_s$

2.7 References

- [1] T.P. Wangler, *RF Linear Accelerators*, 2nd ed. (Wiley-VCH, 2008), Ch. 6.
- [2] S.Y. Lee, *Accelerator Physics*, 3rd ed. (World Scientific, 2012), Ch. 3, Sec. VIII.4
- [3] H. Wiedemann, *Particle Accelerator Physics*, 4th ed. (Springer, 2015), Ch. 9

Chapter 3

Longitudinal Focusing and Adiabatic Damping

3.1 Introduction

In the previous note we derived the longitudinal equations of motion and the Hamiltonian that generates them. Here we develop the physics that follows: phase stability, focusing strength, the acceleration–focusing trade-off, fixed points, separatrix geometry, and the adiabatic damping that occurs as the beam gains energy.

For self-containedness we restate the equations of motion and Hamiltonian. The canonical pair is $(\Delta\varphi, \Delta W)$ with independent variable s :

$$\begin{aligned}\frac{d\Delta\varphi}{ds} &= -\frac{2\pi}{\lambda m_0 c^2 \beta_s^3 \gamma_s^3} \Delta W \\ \frac{d\Delta W}{ds} &= qE_0 T [\cos(\varphi_s + \Delta\varphi) - \cos \varphi_s] \\ H &= -\frac{\pi}{\lambda m_0 c^2 \beta_s^3 \gamma_s^3} \Delta W^2 - qE_0 T [\sin(\varphi_s + \Delta\varphi) - \Delta\varphi \cos \varphi_s]\end{aligned}$$

When $\beta_s \gamma_s$ and $E_0 T$ are constant (or slowly varying), H is conserved and phase-space trajectories are level curves of H .

3.2 Phase Stability

Linearize the energy equation about the synchronous particle ($\Delta\varphi \ll 1$):

$$\cos(\varphi_s + \Delta\varphi) - \cos \varphi_s \approx -\sin \varphi_s \Delta\varphi$$

The linearized system is

$$\frac{d\Delta\varphi}{ds} = -\frac{2\pi}{\lambda m_0 c^2 \beta_s^3 \gamma_s^3} \Delta W, \quad \frac{d\Delta W}{ds} = -qE_0 T \sin \varphi_s \Delta\varphi$$

Differentiating the first and substituting the second:

$$\frac{d^2 \Delta\varphi}{ds^2} = \frac{2\pi q E_0 T \sin \varphi_s}{\lambda m_0 c^2 \beta_s^3 \gamma_s^3} \Delta\varphi$$

This is a harmonic oscillator if and only if the coefficient of $\Delta\varphi$ is negative:

$$\sin \varphi_s < 0 \quad \Longleftrightarrow \quad -\pi < \varphi_s < 0$$

This is the **phase stability condition**: the synchronous phase must lie on the falling slope of the RF wave. Consider a particle that arrives late ($\Delta\varphi > 0$). With $\sin \varphi_s < 0$, the linearized energy equation gives $d\Delta W/ds > 0$ — the particle gains more energy than the synchronous particle and speeds up, restoring it toward the design phase. An early particle ($\Delta\varphi < 0$) gains less energy and slows down. This is the restoring mechanism.

Additionally, the synchronous particle must gain energy. The energy gain per gap is proportional to $\cos \varphi_s$, so net acceleration requires:

$$\cos \varphi_s > 0 \quad \Longleftrightarrow \quad -\frac{\pi}{2} < \varphi_s < \frac{\pi}{2}$$

Combining stability and acceleration:

$$\boxed{-\frac{\pi}{2} < \varphi_s < 0}$$

3.3 Longitudinal Focusing Strength

From the linearized second-order equation we identify the synchrotron wavenumber:

$$\frac{d^2 \Delta\varphi}{ds^2} + k_s^2 \Delta\varphi = 0$$

$$\boxed{k_s^2 = \frac{2\pi q E_0 T (-\sin \varphi_s)}{\lambda m_0 c^2 \beta_s^3 \gamma_s^3}}$$

The synchrotron wavelength $\lambda_s = 2\pi/k_s$ gives the distance over which the particle completes one full oscillation. The number of cells per oscillation period is $N_{\text{osc}} = \lambda_s/L_{\text{cell}}$. For the smooth approximation (differential equations instead of difference equations) to hold, we need $N_{\text{osc}} \gg 1$. In typical linacs, $N_{\text{osc}} \sim 50\text{--}200$.

The focusing strength ($-\sin \varphi_s$) and acceleration ($\cos \varphi_s$) depend on φ_s :

φ_s	$-\sin \varphi_s$	$\cos \varphi_s$	Character
0	0	1	Maximum acceleration, no focusing
-30°	0.5	0.87	Moderate focusing, strong acceleration
-45°	0.71	0.71	Equal focusing and acceleration
-60°	0.87	0.5	Strong focusing, moderate acceleration
-90°	1	0	Maximum focusing, zero net acceleration

At $\varphi_s = 0$, particles accelerate maximally but there is no restoring force. At $\varphi_s = -\pi/2$, focusing is maximized but the beam gains no energy.

3.4 Acceleration vs. Focusing Trade-off

The two performance metrics are:

- **Acceleration:** energy gain per unit length $\propto \cos \varphi_s$
- **Focusing:** restoring force $\propto (-\sin \varphi_s)$

Since $\sin^2 \varphi_s + \cos^2 \varphi_s = 1$, one cannot simultaneously maximize both. This is the fundamental constraint governing the choice of synchronous phase.

3.5 Fixed Points

A fixed point is a location in $(\Delta\varphi, \Delta W)$ space where both derivatives vanish simultaneously. From the equations of motion:

- $d\Delta\varphi/ds = 0$ requires $\Delta W = 0$
- $d\Delta W/ds = 0$ requires $\cos(\varphi_s + \Delta\varphi) = \cos \varphi_s$

The second condition has two solutions in $[-\pi, \pi]$:

Stable fixed point (SFP): $(\Delta\varphi, \Delta W) = (0, 0)$. The Jacobian eigenvalues are purely imaginary ($\pm ik_s$) — nearby trajectories form closed orbits. This is a center.

Unstable fixed point (UFP): $(\Delta\varphi, \Delta W) = (-2\varphi_s, 0)$. The eigenvalues are real ($\pm|k_s|$) — nearby trajectories diverge hyperbolically. This is a saddle. The separatrix passes through this point.

3.6 Separatrix and Bucket Area

3.6.1 Derivation

Since H is conserved, orbits are level curves $H(\Delta\varphi, \Delta W) = \text{const.}$ The separatrix is the level curve passing through the UFP:

$$H(\Delta\varphi, \Delta W) = H(-2\varphi_s, 0)$$

Evaluate H at the UFP. Setting $\Delta\varphi = -2\varphi_s$ and $\Delta W = 0$:

$$H_{\text{UFP}} = -qE_0T [\sin(\varphi_s - 2\varphi_s) - (-2\varphi_s) \cos \varphi_s] = -qE_0T [-\sin \varphi_s + 2\varphi_s \cos \varphi_s]$$

Set $H = H_{\text{UFP}}$ and solve for ΔW^2 . Writing out both sides:

$$-\frac{\pi}{\lambda m_0 c^2 \beta_s^3 \gamma_s^3} \Delta W^2 - qE_0T [\sin(\varphi_s + \Delta\varphi) - \Delta\varphi \cos \varphi_s] = -qE_0T [-\sin \varphi_s + 2\varphi_s \cos \varphi_s]$$

Move the potential term to the right:

$$-\frac{\pi}{\lambda m_0 c^2 \beta_s^3 \gamma_s^3} \Delta W^2 = -qE_0T [-\sin \varphi_s + 2\varphi_s \cos \varphi_s] + qE_0T [\sin(\varphi_s + \Delta\varphi) - \Delta\varphi \cos \varphi_s]$$

Factor out qE_0T on the right:

$$-\frac{\pi}{\lambda m_0 c^2 \beta_s^3 \gamma_s^3} \Delta W^2 = qE_0T [\sin(\varphi_s + \Delta\varphi) + \sin \varphi_s - (2\varphi_s + \Delta\varphi) \cos \varphi_s]$$

Multiply both sides by $-\lambda m_0 c^2 \beta_s^3 \gamma_s^3 / \pi$:

$$\Delta W^2 = \frac{\lambda m_0 c^2 \beta_s^3 \gamma_s^3 q E_0 T}{\pi} [(2\varphi_s + \Delta\varphi) \cos \varphi_s - \sin(\varphi_s + \Delta\varphi) - \sin \varphi_s]$$

Verification at the UFP: set $\Delta\varphi = -2\varphi_s$. The bracket becomes

$$(2\varphi_s - 2\varphi_s) \cos \varphi_s - \sin(-\varphi_s) - \sin \varphi_s = 0 + \sin \varphi_s - \sin \varphi_s = 0 \quad \checkmark$$

The separatrix formula is therefore

$$\Delta W = \pm \sqrt{\frac{\lambda m_0 c^2 \beta_s^3 \gamma_s^3 q E_0 T}{\pi} [(2\varphi_s + \Delta\varphi) \cos \varphi_s - \sin(\varphi_s + \Delta\varphi) - \sin \varphi_s]}$$

3.6.2 Bucket height

The maximum $|\Delta W|$ occurs at $\Delta\varphi = 0$. Evaluating the bracket:

$$(2\varphi_s) \cos \varphi_s - \sin \varphi_s - \sin \varphi_s = 2\varphi_s \cos \varphi_s - 2 \sin \varphi_s = 2(\varphi_s \cos \varphi_s - \sin \varphi_s)$$

Hence

$$\Delta W_{\max}^2 = \frac{2\lambda m_0 c^2 \beta_s^3 \gamma_s^3 q E_0 T}{\pi} (\varphi_s \cos \varphi_s - \sin \varphi_s)$$

For $-\pi/2 < \varphi_s < 0$: $\varphi_s \cos \varphi_s < 0$ and $-\sin \varphi_s > 0$, so the sum $(\varphi_s \cos \varphi_s - \sin \varphi_s) > 0$ and $\Delta W_{\max}$ is real. This matches Wangler Eq. 6.28.

3.6.3 Bucket length

The bucket extends along the $\Delta W = 0$ axis between the two zeros of the bracket. One zero is the UFP at $\Delta\varphi = -2\varphi_s$. The other zero depends on φ_s and must generally be found numerically.

3.6.4 Physical meaning

Particles inside the separatrix execute bounded oscillations about the synchronous particle — they are **captured** and continue to be accelerated. Particles outside the separatrix slip continuously in phase and gain no net energy — they are **uncaptured**. The area enclosed by the separatrix defines the longitudinal acceptance of the linac.

3.6.5 Plotting the separatrix

From the boxed formula, plot both branches (+ and −) over the $\Delta\varphi$ range where the expression under the square root is non-negative. One boundary is the UFP at $\Delta\varphi = -2\varphi_s$; the other must be found numerically.

3.6.6 Dependence on φ_s

As $|\varphi_s| \rightarrow 0$: the UFP at $\Delta\varphi = -2\varphi_s$ merges with the SFP, the bucket collapses, and the restoring force vanishes — the degenerate case of zero focusing. As $|\varphi_s| \rightarrow \pi/2$: the bucket expands toward the full 2π phase range (UFP moves to $\Delta\varphi = \pi$) and focusing is maximized, but acceleration vanishes. Intermediate values balance acceptance against acceleration.

3.7 Adiabatic Damping

3.7.1 Linearized Hamiltonian near the stable fixed point

Expand the potential in H for small oscillations about $(\Delta\varphi, \Delta W) = (0, 0)$. Using $\sin(\varphi_s + \Delta\varphi) \approx \sin \varphi_s + \cos \varphi_s \Delta\varphi$:

$$\sin(\varphi_s + \Delta\varphi) - \Delta\varphi \cos \varphi_s \approx \sin \varphi_s + \cos \varphi_s \Delta\varphi - \Delta\varphi \cos \varphi_s = \sin \varphi_s$$

This zeroth-order piece is a constant and does not contribute to the dynamics. We need the next order. Retaining the $\Delta\varphi^2$ term in the Taylor expansion $\sin(\varphi_s + \Delta\varphi) \approx \sin \varphi_s + \cos \varphi_s \Delta\varphi - \frac{1}{2} \sin \varphi_s \Delta\varphi^2$:

$$\sin(\varphi_s + \Delta\varphi) - \Delta\varphi \cos \varphi_s \approx \sin \varphi_s - \frac{1}{2} \sin \varphi_s \Delta\varphi^2$$

Dropping the constant $\sin \varphi_s$ (it shifts H but not the equations of motion), the linearized Hamiltonian is

$$H_{\text{lin}} = -\frac{\pi}{\lambda m_0 c^2 \beta_s^3 \gamma_s^3} \Delta W^2 + \frac{1}{2} q E_0 T \sin \varphi_s \Delta\varphi^2$$

Since $\sin \varphi_s < 0$ in the stability range, write $(-\sin \varphi_s) > 0$ and define the shorthand

$$A \equiv \frac{2\pi}{\lambda m_0 c^2 \beta_s^3 \gamma_s^3}, \quad B \equiv q E_0 T (-\sin \varphi_s)$$

so that

$$H_{\text{lin}} = -\frac{A}{2} \Delta W^2 - \frac{B}{2} \Delta\varphi^2$$

(Both terms are negative because the sign convention of our Hamiltonian has $d\Delta\varphi/ds = \partial H/\partial(\Delta W)$.) Hamilton's equations reproduce the linearized system:

$$\frac{d\Delta\varphi}{ds} = \frac{\partial H_{\text{lin}}}{\partial(\Delta W)} = -A \Delta W, \quad \frac{d\Delta W}{ds} = -\frac{\partial H_{\text{lin}}}{\partial(\Delta\varphi)} = B \Delta\varphi$$

Eliminating ΔW gives

$$\frac{d^2 \Delta\varphi}{ds^2} + AB \Delta\varphi = 0$$

confirming the synchrotron wavenumber $k_s^2 = AB = \frac{2\pi q E_0 T (-\sin \varphi_s)}{\lambda m_0 c^2 \beta_s^3 \gamma_s^3}$ as derived earlier.

3.7.2 Action variable

For a harmonic oscillator the motion is an ellipse. Using the solution $\Delta\varphi = \hat{\Delta\varphi} \cos(k_s s + \theta_0)$, the conjugate momentum follows from the first equation of motion:

$$\Delta W = -\frac{1}{A} \frac{d\Delta\varphi}{ds} = \frac{k_s}{A} \hat{\Delta\varphi} \sin(k_s s + \theta_0)$$

Hence the energy amplitude is $\Delta\hat{W} = \frac{k_s}{A} \hat{\Delta\varphi}$, and the ratio of semi-axes is

$$\frac{\Delta\hat{W}}{\Delta\hat{\varphi}} = \frac{k_s}{A} = \frac{\sqrt{AB}}{A} = \sqrt{\frac{B}{A}}$$

The action is defined as

$$J = \frac{1}{2\pi} \oint \Delta W \, d(\Delta\varphi)$$

For an ellipse with semi-axes $\Delta\hat{\varphi}$ and $\Delta\hat{W}$, the enclosed area is $\pi \Delta\hat{\varphi} \Delta\hat{W}$, so

$$J = \frac{1}{2\pi} \cdot \pi \Delta\hat{\varphi} \Delta\hat{W} = \frac{1}{2} \Delta\hat{\varphi} \Delta\hat{W}$$

3.7.3 Adiabatic invariance

In a real linac $\beta_s \gamma_s$ increases with s , so A and B (and hence k_s) vary from cell to cell. If these parameters change slowly compared to the oscillation period — i.e. the fractional change per synchrotron wavelength is small — then the action is an adiabatic invariant:

$$J = \frac{1}{2} \Delta\hat{\varphi} \Delta\hat{W} = \text{const}$$

From the semi-axis ratio $\Delta\hat{W}/\Delta\hat{\varphi} = \sqrt{B/A}$ we can express each amplitude in terms of J :

$$\Delta\hat{\varphi} = \sqrt{\frac{2J}{\sqrt{B/A}}} = \sqrt{2J} \left(\frac{A}{B}\right)^{1/4}, \quad \Delta\hat{W} = \sqrt{2J} \left(\frac{B}{A}\right)^{1/4}$$

Now extract the $\beta_s \gamma_s$ dependence. The RF field $E_0 T$ and λ are properties of the structure and change slowly; the dominant s -dependence enters through $A \propto (\beta_s^3 \gamma_s^3)^{-1}$. Meanwhile $B = q E_0 T (-\sin \varphi_s)$ is essentially constant for fixed φ_s and slowly varying $E_0 T$. Therefore

$$\frac{A}{B} \propto (\beta_s^3 \gamma_s^3)^{-1}$$

and the amplitudes scale as

$$\Delta\hat{\varphi} \propto [(\beta_s^3 \gamma_s^3)^{-1}]^{1/4} = (\beta_s \gamma_s)^{-3/4}$$

$$\Delta\hat{W} \propto [(\beta_s^3 \gamma_s^3)^{-1}]^{-1/4} = (\beta_s \gamma_s)^{3/4}$$

$$\boxed{\Delta\varphi_{\max} \propto (\beta_s \gamma_s)^{-3/4}, \quad \Delta W_{\max} \propto (\beta_s \gamma_s)^{3/4}}$$

3.7.4 Interpretation

As the beam accelerates, the phase amplitude $\Delta\hat{\varphi}$ decreases — the bunch gets shorter. This is called **adiabatic damping** of the phase oscillation. The energy amplitude $\Delta\hat{W}$ simultaneously grows, and the phase-space area $\pi \Delta\hat{\varphi} \Delta\hat{W} = 2\pi J = \text{const}$ is conserved throughout — Liouville's theorem is satisfied. There is no dissipation; the ellipse changes aspect ratio but not area.

3.8 Summary

Phase stability:

$$-\frac{\pi}{2} < \varphi_s < 0$$

Synchrotron wavenumber:

$$k_s^2 = \frac{2\pi q E_0 T (-\sin \varphi_s)}{\lambda m_0 c^2 \beta_s^3 \gamma_s^3}$$

Adiabatic amplitude scaling:

$$\Delta\varphi_{\max} \propto (\beta_s \gamma_s)^{-3/4}, \quad \Delta W_{\max} \propto (\beta_s \gamma_s)^{3/4}$$

Design parameter	Controlled by φ_s
Stability	$\sin \varphi_s < 0$ (phase on falling slope)
Focusing strength	$\propto -\sin \varphi_s$ (maximized at -90°)
Acceleration	$\propto \cos \varphi_s$ (maximized at 0°)
Bucket area	Larger for larger $ \varphi_s $; vanishes as $\varphi_s \rightarrow 0$

3.9 Appendix: Cosine vs Sine Convention

Quantity	Cosine (φ)	Sine (ϕ)
Phase variable	φ	$\phi = \varphi + \pi/2$
Energy gain	$\Delta W = q E_0 T L \cos \varphi_s$	$\Delta W = q E_0 T L \sin \phi_s$
Stable range (accel + focusing)	$-\pi/2 < \varphi_s < 0$	$0 < \phi_s < \pi/2$
Phase stability slope condition	$\sin \varphi_s < 0$	$\cos \phi_s > 0$
Focusing strength	$-\sin \varphi_s$	$\cos \phi_s$
Acceleration coefficient	$\cos \varphi_s$	$\sin \phi_s$
Maximum focusing	$\varphi_s = -\pi/2$ (no accel)	$\phi_s = 0$ (no accel)
Maximum acceleration	$\varphi_s = 0$ (no focusing)	$\phi_s = \pi/2$ (no focusing)
Synchrotron wavenumber k_s^2	$\frac{-2\pi q E_0 T \sin \varphi_s}{\lambda m_0 c^2 \beta_s^3 \gamma_s^3}$	$\frac{2\pi q E_0 T \cos \phi_s}{\lambda m_0 c^2 \beta_s^3 \gamma_s^3}$
SFP	$(\Delta\varphi, \Delta W) = (0, 0)$	$(\Delta\phi, \Delta W) = (0, 0)$
UFP	$(-2\varphi_s, 0)$	$(\pi - 2\phi_s, 0)$
Separatrix $\Delta W^2 =$	$\frac{\lambda m_0 c^2 \beta_s^3 \gamma_s^3 q E_0 T}{\pi} [(2\varphi_s + \Delta\varphi) \cos \varphi_s - \sin(\varphi_s + \Delta\varphi) - \sin \varphi_s]$	$\frac{\lambda m_0 c^2 \beta_s^3 \gamma_s^3 q E_0 T}{\pi} [(2\phi_s - \pi + \Delta\phi) \sin \phi_s + \cos(\phi_s + \Delta\phi) + \cos \phi_s]$
Bucket height $\Delta W_{\max}^2 \propto$	$\varphi_s \cos \varphi_s - \sin \varphi_s$	$(\phi_s - \pi/2) \sin \phi_s + \cos \phi_s$

3.10 References

- [1] T.P. Wangler, *RF Linear Accelerators*, 2nd ed. (Wiley-VCH, 2008), Ch. 6.
- [2] S.Y. Lee, *Accelerator Physics*, 3rd ed. (World Scientific, 2012), Ch. 3, Sec. II.
- [3] H. Wiedemann, *Particle Accelerator Physics*, 4th ed. (Springer, 2015), Ch. 9.

Chapter 4

RF Defocusing

4.1 Motivation from the Longitudinal Hamiltonian

The longitudinal Hamiltonian derived in the previous note is

$$H = -\frac{\pi}{\lambda m_0 c^2 \beta_s^3 \gamma_s^3} \Delta W^2 - q E_0 T [\sin(\varphi_s + \Delta\varphi) - \Delta\varphi \cos \varphi_s]$$

The potential term encodes phase stability: particles oscillate in the RF bucket provided $-\pi/2 < \varphi_s < 0$. This dynamics is generated by an on-axis longitudinal field

$$E_z(0, z, t) = E_0(z) \cos(\omega t + \varphi)$$

But E_z does not exist in isolation. Maxwell's equations in a source-free region couple longitudinal and transverse field components: any spatial variation of E_z necessarily produces radial electric and azimuthal magnetic fields. These transverse fields exert a radial force on the beam. We now derive these components and show that they are **defocusing** for exactly the phase range that provides longitudinal stability — a fundamental conflict in linac design.

4.2 Deriving Radial Field Components from Maxwell's Equations

4.2.1 Setup

Consider a cylindrically symmetric accelerating structure. We work in cylindrical coordinates (r, θ, z) with $\partial/\partial\theta = 0$ (azimuthal symmetry). The region near the axis is source-free ($\rho = 0$, $\mathbf{J} = 0$).

The accelerating field on axis is

$$E_z(0, z, t) = E_0(z) \cos(\omega t + \varphi)$$

We seek the off-axis fields E_r and B_θ that are required by Maxwell's equations.

4.2.2 Radial electric field from Gauss's law

In a source-free region, $\nabla \cdot \mathbf{E} = 0$:

$$\frac{1}{r} \frac{\partial(r E_r)}{\partial r} + \frac{\partial E_z}{\partial z} = 0$$

(The $\partial E_\theta / \partial \theta$ term vanishes by azimuthal symmetry.)

Rearranging:

$$\frac{\partial(rE_r)}{\partial r} = -r \frac{\partial E_z}{\partial z}$$

Integrating from $r = 0$ (where $E_r = 0$ by symmetry) to a small radius r :

$$rE_r = - \int_0^r r' \frac{\partial E_z(r', z, t)}{\partial z} dr'$$

Near the axis, E_z varies slowly with r (the leading correction is $\mathcal{O}(r^2)$), so we approximate $E_z(r', z, t) \approx E_z(0, z, t)$ inside the integral:

$$rE_r \approx - \frac{\partial E_z(0, z, t)}{\partial z} \int_0^r r' dr' = - \frac{\partial E_z}{\partial z} \cdot \frac{r^2}{2}$$

Therefore:

$$\boxed{E_r = - \frac{r}{2} \frac{\partial E_z}{\partial z}}$$

4.2.3 Azimuthal magnetic field from Ampère–Maxwell’s law

In vacuum, $\nabla \times \mathbf{B} = (1/c^2) \partial \mathbf{E} / \partial t$. The $\hat{z}$ component of the curl gives (for azimuthal symmetry):

$$\frac{1}{r} \frac{\partial(rB_\theta)}{\partial r} = \frac{1}{c^2} \frac{\partial E_z}{\partial t}$$

Integrating as before:

$$rB_\theta = \frac{1}{c^2} \frac{\partial E_z(0, z, t)}{\partial t} \cdot \frac{r^2}{2}$$

Therefore:

$$\boxed{B_\theta = \frac{r}{2c^2} \frac{\partial E_z}{\partial t}}$$

4.2.4 Radial Lorentz force

The total radial force on a particle with charge q and axial velocity $v_z \approx \beta_s c$ is

$$F_r = q(E_r - v_z B_\theta)$$

The minus sign arises from the cross product $v_z \hat{z} \times B_\theta \hat{\theta} = -v_z B_\theta \hat{r}$. Substituting:

$$F_r = q \left(-\frac{r}{2} \frac{\partial E_z}{\partial z} - \frac{\beta_s c r}{2c^2} \frac{\partial E_z}{\partial t} \right) = -\frac{qr}{2} \left(\frac{\partial E_z}{\partial z} + \frac{\beta_s}{c} \frac{\partial E_z}{\partial t} \right)$$

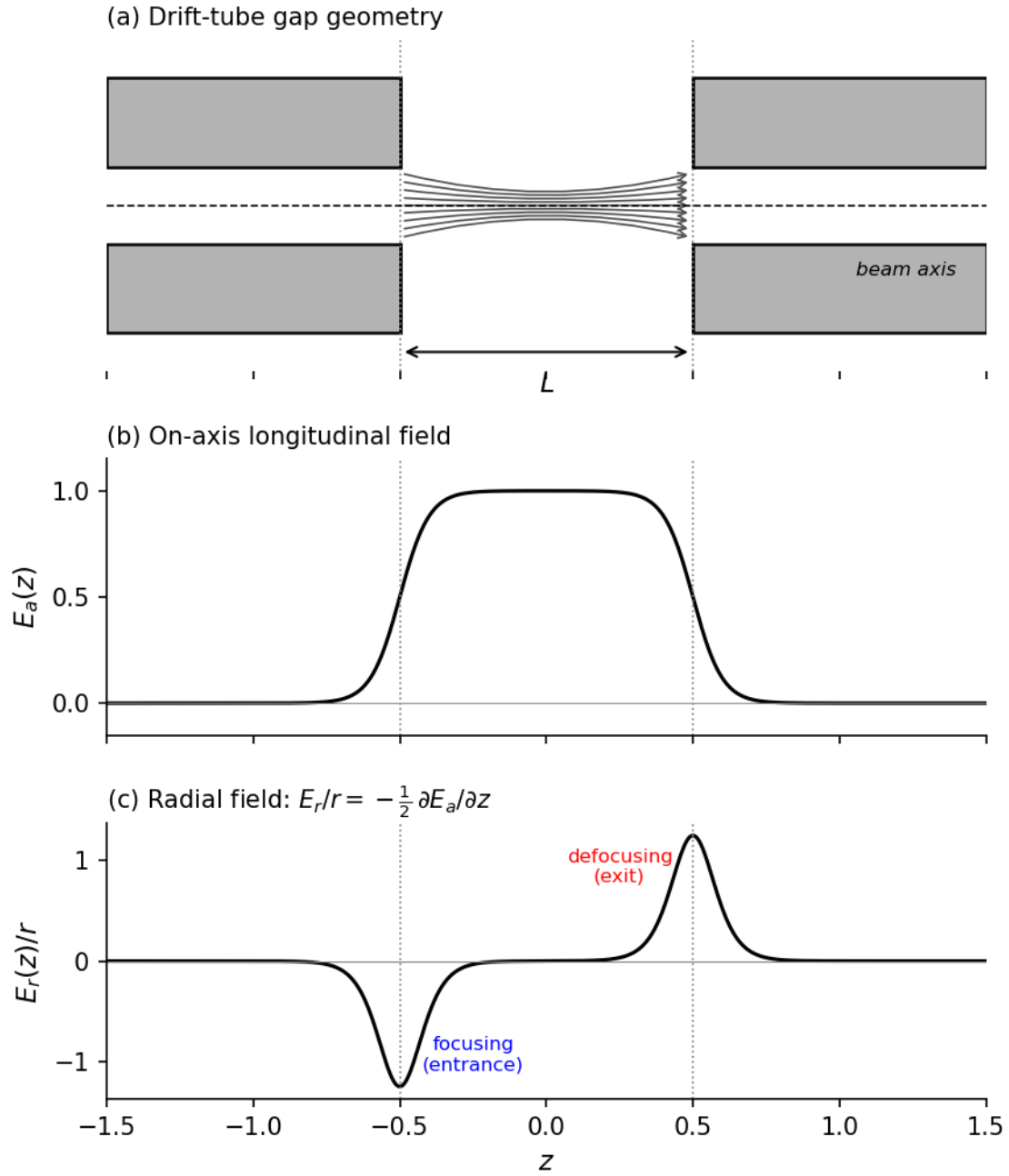


Figure 4.1: RF gap geometry and field profiles. (a) Drift-tube gap with electric field lines bowing outward off-axis. (b) On-axis longitudinal field $E_a(z)$. (c) Radial field per unit radius $E_r/r = -(1/2) \partial E_a / \partial z$: negative (focusing) at the entrance, positive (defocusing) at the exit.

4.3 Physical Picture: Focusing and Defocusing at a Gap

Before computing the net impulse, consider what happens qualitatively as a particle traverses a single RF gap.

The radial electric field $E_r = -(r/2) \partial E_z / \partial z$ depends on the **spatial gradient** of the longitudinal field. In a typical gap:

- **Entering the gap** ($\partial E_z / \partial z > 0$, field is rising): $E_r < 0$, pointing inward → **focusing**
- **Exiting the gap** ($\partial E_z / \partial z < 0$, field is falling): $E_r > 0$, pointing outward → **defocusing**

If the transit time were zero, these two impulses would exactly cancel. But the particle takes a finite time to cross the gap, and during that time the RF field is changing. A particle arriving on the rising slope of the RF wave ($\sin \varphi < 0$) sees the field weaker when entering and stronger when exiting — the defocusing impulse wins. This is the origin of net RF defocusing.

4.4 Net Radial Impulse Through an RF Gap

4.4.1 Setting up the integral

Write the on-axis field as $E_z(0, s, t) = E_0(s) \cos \varphi(t)$ where $\varphi(t) = \omega t + \varphi_0$ is the RF phase seen by the particle at time t . The particle traverses the gap along the axis (s coincides with the axial coordinate z), arriving at position s at time $t(s)$. The radial impulse accumulated over the full gap is (using $dt = ds/(\beta_s c)$):

$$\Delta p_r = \int F_r dt = -\frac{qr}{2\beta_s c} \int_{-L/2}^{L/2} \left(\frac{\partial E_z}{\partial s} + \frac{\beta_s}{c} \frac{\partial E_z}{\partial t} \right) ds$$

where we approximate $r \approx \text{const}$ through the thin gap (paraxial approximation).

4.4.2 Combining the two terms

The total derivative of E_z along the particle trajectory with respect to s is

$$\frac{dE_z}{ds} = \frac{\partial E_z}{\partial s} + \frac{1}{\beta_s c} \frac{\partial E_z}{\partial t}$$

Therefore:

$$\frac{\partial E_z}{\partial s} + \frac{\beta_s}{c} \frac{\partial E_z}{\partial t} = \frac{\partial E_z}{\partial s} + \beta_s^2 \left(\frac{dE_z}{ds} - \frac{\partial E_z}{\partial s} \right) = \frac{1}{\gamma_s^2} \frac{\partial E_z}{\partial s} + \beta_s^2 \frac{dE_z}{ds}$$

The second term is a total derivative; its integral over the gap gives the boundary values of E_z , which vanish because the field is localized. Thus:

$$\Delta p_r = -\frac{qr}{2\gamma_s^2 \beta_s c} \int_{-L/2}^{L/2} \frac{\partial E_z(0, s, t(s))}{\partial s} ds$$

4.4.3 Evaluating with the transit-time factor

For a standing-wave cavity, $E_z(0, s, t) = E_0(s) \cos(\omega t + \varphi_0)$. The spatial partial derivative at fixed t is simply $\partial E_z / \partial s = E_0'(s) \cos(\omega t + \varphi_0)$. Evaluating along the particle trajectory $t(s) = s/(\beta_s c) + t_0$ where the phase at the electrical center is φ_s :

$$\left. \frac{\partial E_z}{\partial s} \right|_{t=t(s)} = E'_0(s) \cos(ks + \varphi_s)$$

where $k = \omega/(\beta_s c)$. Integrating by parts:

$$\int_{-L/2}^{L/2} E'_0(s) \cos(ks + \varphi_s) ds = \left[E_0(s) \cos(ks + \varphi_s) \right]_{-L/2}^{L/2} + k \int_{-L/2}^{L/2} E_0(s) \sin(ks + \varphi_s) ds$$

The boundary term vanishes because $E_0(\pm L/2) = 0$ (field is localized). Expanding $\sin(ks + \varphi_s) = \sin ks \cos \varphi_s + \cos ks \sin \varphi_s$ and using the symmetry of the gap ($E_0(s)$ is even, so the $\sin ks$ integral vanishes):

$$\int_{-L/2}^{L/2} \frac{\partial E_z}{\partial s} ds = k \sin \varphi_s \int_{-L/2}^{L/2} E_0(s) \cos ks ds = \frac{\omega}{\beta_s c} E_0 L T \sin \varphi_s$$

where E_0 is the average field, L is the gap length, and T is the transit-time factor. Therefore:

$$\Delta p_r = -\frac{qr}{2\gamma_s^2 \beta_s c} \cdot \frac{\omega}{\beta_s c} \cdot E_0 L T \sin \varphi_s = -\frac{qr \omega E_0 L T \sin \varphi_s}{2\gamma_s^2 \beta_s^2 c^2}$$

Substituting $\omega = 2\pi c/\lambda$:

$$\Delta p_r = -\frac{\pi q E_0 L T \sin \varphi_s}{\lambda \gamma_s^2 \beta_s^2 c} r$$

The impulse is proportional to r — the RF gap acts as a thin lens. This matches Wangler Eq. (7.8).

4.5 RF Defocusing Condition

4.5.1 The fundamental conflict

The phase stability condition derived in the longitudinal dynamics note requires

$$-\frac{\pi}{2} < \varphi_s < 0 \quad \implies \quad \sin \varphi_s < 0$$

Substituting into the radial impulse:

$$\Delta p_r = -\frac{\pi q E_0 L T \sin \varphi_s}{\lambda \gamma_s^2 \beta_s^2 c} r$$

Since $\sin \varphi_s < 0$, the factor $-\sin \varphi_s > 0$. For a particle at $r > 0$: $\Delta p_r > 0$ — the particle gains **outward** radial momentum. This is **defocusing**.

$$\text{Longitudinal stability } (\sin \varphi_s < 0) \implies \text{transverse RF defocusing } (\Delta p_r > 0 \text{ for } r > 0)$$

This is the **fundamental constraint** of linac design: the same RF phase that provides longitudinal focusing simultaneously defocuses the beam transversely. External focusing elements (solenoids, quadrupoles) must be provided to overcome the RF defocusing.

4.6 Equivalent Defocusing Wavenumber

4.6.1 From radial impulse to smooth-approximation wavenumber

The radial impulse per gap is $\Delta p_r \propto r$, so we can write the corresponding change in the normalized slope $\gamma_s \beta_s r'$:

$$\Delta(\gamma_s \beta_s r') = \frac{\Delta p_r}{m_0 c} = -\frac{\pi q E_0 L T \sin \varphi_s}{m_0 c^2 \lambda \gamma_s^2 \beta_s^2} r$$

In the smooth approximation (Wangler Eq. 7.11), this kick is smeared over one cell of length L . From the EOM derived below (with $x'' + \dots = 0$), the RF defocusing contribution per unit length is $\Delta(\gamma_s \beta_s r')/(L \cdot \gamma_s \beta_s) = -k_{\text{RF}}^2 r$. Solving for k_{RF}^2 :

$$k_{\text{RF}}^2 = \frac{\pi q E_0 T |\sin \varphi_s|}{m_0 c^2 \lambda (\beta_s \gamma_s)^3}$$

where we used $|\sin \varphi_s| = \sin(-\varphi_s)$ and $E_0 L/L = E_0$. This matches Wangler Eq. (7.12): $k_{i0}^2 = 2\pi q E_0 T \sin(-\phi)/[mc^2 \lambda (\gamma\beta)^3]$, with the factor of 2 arising because Wangler's EOM (Eq. 7.11) has a coefficient $k_{i0}^2/2$ in front of r , not k_{i0}^2 .

$$k_{\text{RF}}^2 = \frac{\pi q E_0 T |\sin \varphi_s|}{m_0 c^2 \lambda (\beta_s \gamma_s)^3}$$

4.6.2 Comparison with external focusing

From the transverse adiabatic damping note, the focusing wavenumber for magnetic quadrupoles with gradient G is

$$k_{\text{ext}}^2 = \frac{|qG|}{m_0 c \beta_s \gamma_s}$$

For net transverse stability, the external focusing must overcome the RF defocusing:

$$k_{\text{ext}}^2 > k_{\text{RF}}^2$$

The ratio

$$\frac{k_{\text{RF}}^2}{k_{\text{ext}}^2} = \frac{\pi E_0 T |\sin \varphi_s|}{c |G| \lambda (\beta_s \gamma_s)^2}$$

grows at low energy (small $\beta_s \gamma_s$): RF defocusing dominates when the beam is slow. At high energy, external focusing easily wins.

4.7 Design Implications

4.7.1 Energy dependence

The RF defocusing strength scales as $k_{\text{RF}}^2 \propto (\beta_s \gamma_s)^{-3}$, while magnetic focusing scales as $k_{\text{ext}}^2 \propto (\beta_s \gamma_s)^{-1}$. The ratio:

$$\frac{k_{\text{RF}}^2}{k_{\text{ext}}^2} \propto \frac{1}{(\beta_s \gamma_s)^2}$$

At low energy ($\beta_s \ll 1$, $\gamma_s \approx 1$): - The RF defocusing is at its strongest — the $(\beta_s \gamma_s)^3$ denominator in the kick formula is small - External focusing must be correspondingly strong and closely spaced

As the beam accelerates ($\beta_s \gamma_s$ increases): - RF defocusing weakens rapidly - External focusing requirements relax — quadrupole spacing can increase, gradients can decrease

4.7.2 Practical consequences

This energy dependence explains the structure of real linacs:

Energy regime	β_s	Focusing solution
Very low (< 1 MeV/u)	< 0.05	RFQ: RF electric quadrupole vanes provide continuous focusing
Low (1–100 MeV/u)	0.05–0.4	DTL: permanent-magnet quadrupoles inside every drift tube
Medium (100–1000 MeV/u)	0.4–0.9	CCL/SCL: quadrupole doublets or triplets between multi-cell tanks
High (> 1 GeV/u)	≈ 1	Long FODO periods; RF defocusing negligible

4.8 Transverse Equation of Motion

We now collect all forces — external focusing, RF defocusing, and acceleration (adiabatic damping) — into a single transverse equation of motion.

4.8.1 Starting from first principles

The transverse mechanical momentum is $p_x = m_0 c \beta_s \gamma_s x'$, where primes denote d/ds . Newton's second law in the transverse direction gives $dp_x/dt = F_x$. Converting to the independent variable s using $dt = ds/(\beta_s c)$:

$$\beta_s c \frac{dp_x}{ds} = F_{\text{ext}} + F_{\text{RF}}$$

Expanding the left side:

$$\beta_s c \frac{d}{ds}(m_0 c \beta_s \gamma_s x') = m_0 c^2 \beta_s \left[\beta_s \gamma_s x'' + \frac{d(\beta_s \gamma_s)}{ds} x' \right]$$

Dividing through by $m_0 c^2 \beta_s^2 \gamma_s$:

$$x'' + \frac{1}{\beta_s \gamma_s} \frac{d(\beta_s \gamma_s)}{ds} x' = \frac{F_{\text{ext}} + F_{\text{RF}}}{m_0 c^2 \beta_s^2 \gamma_s}$$

4.8.2 Identifying the damping coefficient

From the identity $\beta\gamma = \sqrt{\gamma^2 - 1}$, the damping coefficient is

$$\frac{1}{\beta_s \gamma_s} \frac{d(\beta_s \gamma_s)}{ds} = \frac{1}{\beta_s^2 \gamma_s} \frac{d\gamma_s}{ds} = \frac{\gamma'_s}{\beta_s^2 \gamma_s}$$

where $\gamma'_s \equiv d\gamma_s/ds$.

4.8.3 Identifying the force terms

For a magnetic quadrupole with gradient G , the Lorentz force is $F_x = -qv_z Gx = -q\beta_s c Gx$. Dividing by $m_0 c^2 \beta_s^2 \gamma_s$:

$$\frac{F_{\text{ext}}}{m_0 c^2 \beta_s^2 \gamma_s} = -\frac{qG}{m_0 c \beta_s \gamma_s} x = -k_{\text{ext}}^2 x$$

confirming $k_{\text{ext}}^2 = |qG|/(m_0 c \beta_s \gamma_s)$ as before. The RF defocusing, from the continuous-force form of the thin-lens result (Wangler Eq. 7.11), contributes:

$$\frac{F_{\text{RF}}}{m_0 c^2 \beta_s^2 \gamma_s} = +k_{\text{RF}}^2 x$$

(positive sign = defocusing), where $k_{\text{RF}}^2 = \pi q E_0 T |\sin \varphi_s| / [m_0 c^2 \lambda (\beta_s \gamma_s)^3]$.

4.8.4 Complete transverse equation of motion

$$x'' + \frac{\gamma'_s}{\beta_s^2 \gamma_s} x' + (k_{\text{ext}}^2 - k_{\text{RF}}^2) x = 0$$

This is the smooth-approximation form of Wangler Eq. (7.46).

4.8.5 Interpretation of each term

Adiabatic damping ($\gamma'_s x' / (\beta_s^2 \gamma_s)$): This is a velocity-dependent “friction” term that damps the oscillation amplitude as the beam accelerates. It is not dissipative — it reflects the shrinking of $x' = p_x / (m_0 c \beta_s \gamma_s)$ at constant p_x as $\beta_s \gamma_s$ grows. This is the same adiabatic damping derived from canonical arguments in the transverse adiabatic damping note; here we see it arise naturally as the convective derivative of the momentum normalization.

Net focusing ($k_{\text{ext}}^2 - k_{\text{RF}}^2$): The effective transverse restoring force is the difference between external focusing and RF defocusing. Stability requires $k_{\text{ext}}^2 > k_{\text{RF}}^2$. The net phase advance per focusing period is

$$\sigma = \sqrt{k_{\text{ext}}^2 - k_{\text{RF}}^2} \cdot L_{\text{period}}$$

which must remain real and below π (alternating-gradient stability limit).

RF defocusing ($k_{\text{RF}}^2 x$): Reconnecting to the derivation above, this term has a definite energy scaling:

$$k_{\text{RF}}^2 = \frac{\pi q E_0 T |\sin \varphi_s|}{m_0 c^2 \lambda (\beta_s \gamma_s)^3}$$

At injection ($\beta_s \approx 0.05$ for a proton DTL), this term is $\sim 10^3$ times larger than at $\beta_s \approx 0.5$. The rapid decrease with energy means that linac focusing lattice design evolves section by section: tight focusing at low energy, relaxed focusing at high energy.

4.9 Summary

The key results of this note:

1. **Maxwell's equations require transverse fields:** $E_r = -(r/2) \partial E_z / \partial z$ and $B_\theta = (r/2c^2) \partial E_z / \partial t$
2. **Net radial impulse per gap:** $\Delta p_r = -\frac{\pi q E_0 L T \sin \varphi_s}{\lambda \gamma_s^2 \beta_s^2 c} r$ (defocusing for $\sin \varphi_s < 0$)
3. **Fundamental conflict:** longitudinal stability ($\sin \varphi_s < 0$) implies transverse defocusing ($\Delta p_r > 0$ for $r > 0$)
4. **Energy scaling:** $k_{\text{RF}}^2 \propto (\beta_s \gamma_s)^{-3}$ — RF defocusing is strongest at low energy
5. **Transverse equation of motion:** $x'' + (\gamma'_s / \beta_s^2 \gamma_s) x' + (k_{\text{ext}}^2 - k_{\text{RF}}^2) x = 0$

4.10 References

[1] T.P. Wangler, *RF Linear Accelerators*, 2nd ed. (Wiley-VCH, 2008), Ch. 7: Sec. 7.1–7.3 (radial impulse, RF defocusing), Eq. (7.11) (transverse EOM with adiabatic damping), Eq. (7.46) (Hill's equation with RF defocusing term), Sec. 7.17 (smooth approximation).

[2] S.Y. Lee, *Accelerator Physics*, 4th ed. (World Scientific, 2019), Ch. 3, Sec. VIII.5 (“Transverse beam dynamics in a linac”), pp. 387–390: Eq. (3.337) (TM₀₁₀ field components), Eq. (3.338) (net transverse defocusing force), Eq. (3.341)–(3.342) (betatron EOM with smooth approximation), Fig. 3.50 (field-line schematic).

4.10.1 Convention comparison: Wangler vs. SY Lee

SY Lee uses the **synchrotron (sine) convention** while Wangler and these notes use the **linac (cosine) convention**. The two are related by $\psi = \varphi + \pi/2$ (see SY Lee preface, p. xi).

Aspect	This note / Wangler (cosine)	SY Lee (sine)
On-axis field	$E_z = E_0 \cos(\omega t + \varphi)$	$E_s = \mathcal{E}_0 \sin \psi$
Phase variable	φ	$\psi = \varphi + \pi/2$
Stability condition	$-\pi/2 < \varphi_s < 0$	$0 < \psi_s < \pi/2$ (below transition)
Defocusing $\propto$	$-\sin \varphi_s > 0$	$\cos \psi_s > 0$ (same quantity)
Radial E-field	$E_r = -\frac{r}{2} \frac{\partial E_z}{\partial z}$	$E_r = +\frac{\omega r}{2v_p} \mathcal{E}_0 \cos \psi$ (TM ₀₁₀ mode)
Defocusing force	$\propto \frac{\omega E_0 T \sin \varphi_s }{\beta_s \gamma_s^2 c} r$	$\propto \frac{\omega \mathcal{E}_0 \cos \psi_s}{2\beta \gamma^2 c} r$ [Eq.(3.338)]